IOT-BASED FISH POND MONITORING & MANAGEMENT SYSTEM

Project Specification Documentation & User Manual

Version 1.0

Designed for Bangladesh Aquaculture

1. Project Overview

The IoT-Based Fish Pond Monitoring and Management System is a smart aquaculture solution designed to address the most critical challenges faced by fish farmers in Bangladesh: sudden fish death caused by deteriorating water quality, lack of continuous monitoring capability, and the absence of automated intervention mechanisms.

This system integrates a network of waterproof sensors with an ESP32 microcontroller to continuously sample water temperature, pH level, and turbidity. All readings are transmitted wirelessly over Wi-Fi using the MQTT protocol to a dedicated Node.js/Express.js backend server, which in turn serves data to a Progressive Web Application (PWA) built with React. The PWA is fully installable on any Android or iOS device — no app store required.

Beyond monitoring, the system actively controls two actuators: an aerator (oxygen pump) and an automated fish feeder, both of which can operate on configurable schedules or be triggered manually via the app.

Project Name	IoT-Based Fish Pond Monitoring & Management System
Document Type	Project Specification & User Manual
Version	1.1
Primary Controller	ESP32 Development Board (30-pin)
Communication	Wi-Fi (IEEE 802.11 b/g/n) + MQTT Protocol
Application Platform	Progressive Web App (React + PWA, installable on Android & iOS)
Backend Stack	Node.js + Express.js + MongoDB + MQTT (Mosquitto)
Authentication	JWT (JSON Web Tokens) + Firebase Auth
Target Environment	Fish Pond / Aquarium (Aquaculture)
Estimated Hardware Cost	4,800 - 5,500 BDT (Base Configuration)
Country / Context	Bangladesh Aquaculture Sector

2. Objectives

- Provide continuous, real-time monitoring of critical water quality parameters including temperature, pH, and turbidity.
- Significantly reduce fish mortality rates by enabling rapid detection of and response to hazardous water conditions.
- Enable remote monitoring and control of the pond ecosystem through a Progressive Web Application installable on any smartphone.
- Implement an intelligent alert system that notifies the user via web push notification, local buzzer, and LED indicator whenever water conditions cross configured safety thresholds.
- Automate the aerator to activate automatically when water temperature exceeds a user-defined threshold, ensuring dissolved oxygen levels remain safe.
- Provide a flexible, configurable automated fish feeding schedule with both time-of-day scheduling and duration control.
- Offer a cost-effective, scalable solution suitable for small-scale aquariums up to large commercial fish ponds.
- Create a foundation for future expansion including AI-based prediction, multi-pond management, solar power integration, and SMS alerting.

3. System Architecture

3.1 High-Level System Overview

The system is built on a four-tier architecture:

- Perception Layer: The sensor array (DS18B20, PH4502C, Turbidity Sensor) continuously samples the physical water environment and delivers analog or digital readings to the ESP32.
- Processing & Connectivity Layer: The ESP32 microcontroller processes all sensor data, applies threshold logic, controls the relay-driven actuators (aerator and feeder motor), drives the local alert system (buzzer and LED), and transmits all data to the backend server over Wi-Fi via the MQTT protocol.
- Backend Layer: A Node.js + Express.js server acts as the central hub. It runs a Mosquitto MQTT broker to receive device messages, stores data in MongoDB, manages user authentication via JWT, and uses Socket.IO to push real-time updates to the frontend.
- Application Layer: A React-based Progressive Web Application connects to the backend via REST API and Socket.IO, displays live sensor readings on a dashboard, allows the user to configure thresholds and schedules, and delivers web push notifications when alerts are triggered.

3.2 Data Flow

- Sensors continuously collect water quality data every configurable interval (default: every 10 seconds).
- The ESP32 reads sensor values, applies calibration offsets, evaluates them against configured thresholds, and publishes data to the MQTT broker topic fishpond/{device_id}/sensors.
- The Node.js backend (subscribed to the MQTT broker) receives the payload, stores it in MongoDB, and broadcasts it to connected PWA clients via Socket.IO in real time.
- The PWA dashboard updates instantly upon receiving the Socket.IO event.
- If a threshold is exceeded, the Node.js backend triggers a Web Push Notification to the registered PWA client. Simultaneously, the ESP32 activates the local buzzer and LED.
- User commands from the PWA (manual actuator control, threshold changes, schedule updates) are sent to the Node.js REST API, which publishes control commands back to the ESP32 via MQTT topic fishpond/{device_id}/commands.

4. Hardware Components

All components listed below have been selected for compatibility, availability in Bangladesh, and optimal price-to-performance ratio. Prices reflect current market rates from verified Bangladeshi suppliers.

Component	Role / Description	Price (BDT)	Source
ESP32 Dev Board (30-pin)	Main controller; handles all sensor reading, Wi-Fi, MQTT, relay control, and app communication	430	roboticsbd.com
DS18B20 (Waterproof)	Measures water temperature with ±0.5°C accuracy; fully waterproof probe for submersion	220	daraz.com.bd
pH Sensor PH4502C	Measures water acidity/alkalinity (0-14 pH range); analog output with onboard signal conditioning	2,900	techshopbd.com

Turbidity Sensor	Detects suspended particles in water; indicates pollution levels and algae growth	820	techshopbd.com
Relay Module (2-channel)	Controls the aerator and feeder motor; switches high-voltage/high-current loads safely from ESP32	400	roboticsbd.com
Buzzer + LED	Local alert system; provides audible and visual alarm when thresholds are breached	200	roboticsbd.com
18650 Battery + TP4056	Rechargeable power supply; portable and suitable for outdoor deployment	100	roboticsbd.com
Aquarium Aerator (small)	Provides oxygen to the water; controlled via relay by ESP32 automatically or manually	~300	Local market
Feeder Motor (DC)	Dispenses fish feed on schedule; driven via relay module from ESP32	~200	Local market
Supporting Components	Breadboard, jumper wires, waterproof enclosure box, 4.7k resistor, capacitors	~200	Various

4.1 ESP32 Development Board

The ESP32 serves as the central intelligence of the entire system. It integrates a dual-core Xtensa LX6 processor running at up to 240 MHz, built-in Wi-Fi (IEEE 802.11 b/g/n), and Bluetooth, making it an ideal choice for IoT applications without requiring any additional wireless communication modules. In this updated architecture, the ESP32 uses the PubSubClient library to communicate with the backend via MQTT.

- Processor: Dual-core Xtensa LX6, up to 240 MHz
- Wi-Fi: IEEE 802.11 b/g/n — enables MQTT communication with the Node.js backend
- ADC: 12-bit Analog-to-Digital Converter for reading pH and turbidity sensors
- GPIO: Multiple digital I/O pins for DS18B20 (one-wire), relay control, buzzer, and LED
- Power: Operates at 3.3V logic with 5V input via USB or VIN pin
- Price: BDT 430 | Alternative: NodeMCU ESP8266 (BDT 350, less processing power)

4.2 DS18B20 Waterproof Temperature Sensor

The DS18B20 uses the one-wire protocol, requiring only a single data pin on the ESP32 for communication. Its fully waterproof stainless steel probe design makes it ideal for direct submersion into pond water.

- Measurement Range: -55°C to +125°C
- Accuracy: ±0.5°C across the operating range
- Interface: Digital (One-Wire protocol)
- Power Supply: 3.0V to 5.5V
- Price: BDT 220

4.3 pH Sensor (PH 4502C Module)

The PH4502C module includes an onboard operational amplifier and signal conditioning circuit that converts the glass electrode potential difference into a readable analog voltage output (0-5V) proportional to the pH level of the water. Calibration using standard buffer solutions (pH 4.0 and pH 7.0) is required for accurate readings.

- pH Measurement Range: 0 to 14
- Operating Temperature: 0 to 60°C

- Output: Analog voltage (0-5V)
- Calibration: Required with standard pH buffer solutions
- Price: BDT 2,900 | Alternative: Gravity pH Sensor (BDT 3,750, more stable)

4.4 Turbidity Sensor

The turbidity sensor uses an infrared LED and phototransistor to measure the amount of light scattered by suspended particles in the water. Higher turbidity results in more light scattering and a higher analog output value.

- Detection: Suspended particles, algae, sediment
- Output: Analog voltage
- Price: BDT 820 | Alternative: Higher-accuracy module at BDT 965

4.5 Relay Module (2-Channel)

The relay module acts as an electrically controlled switch, allowing the ESP32 (3.3V logic) to safely control the aerator and feeder motor. The 2-channel version simultaneously manages both actuators.

- Channels: 2 (one for aerator, one for feeder motor)
- Input Trigger: TTL level (3.3V compatible)
- Output: Controls up to 10A at 250VAC or 10A at 30VDC
- Price: BDT 400 | Alternative: Solid State Relay (faster switching, slightly higher cost)

4.6 Buzzer and LED

The buzzer and LED form the local alert system that provides immediate on-site warning when water quality parameters exceed configured thresholds, independent of internet connectivity.

- Buzzer Type: Mechanical or piezoelectric active buzzer
- LED: Standard 5mm LED (red recommended for alerts)
- Control: Simple digital HIGH/LOW signal from ESP32 GPIO
- Price: BDT 200 (combined)

4.7 Power System

- Primary Option: 18650 Li-ion battery + TP4056 charging module (BDT 100)
- Outdoor Option: Solar panel (5W, 5V) + battery (BDT 500-700) for fully off-grid deployment
- Indoor Test: Standard 5V USB power adapter (no additional cost)

5. Monitored Parameters & Thresholds

The system monitors three core water quality parameters. Default thresholds are pre-configured based on standard aquaculture guidelines for Bangladesh. All thresholds are adjustable by the user through the Progressive Web Application.

Parameter	Normal Range	Alert Threshold	Sensor Used
Water Temperature	25 - 30 °C	Below 20°C / Above 32°C	DS18B20
pH Level	6.5 - 8.5	Below 6.0 / Above 9.0	PH 4502C
Turbidity	Low (Clear Water)	High Turbidity Reading	Turbidity Sensor

Aerator Status	ON / OFF	Auto-ON when temp exceeds limit	Relay Module
Feeder Status	Scheduled / Manual	User-configured timer	Motor + Relay

Note: The above thresholds represent general aquaculture guidelines. Different fish species may require different ranges. Users are advised to consult species-specific requirements and adjust thresholds in the mobile application accordingly.

6. Feature Specification

6.1 Real-Time Water Quality Monitoring

The system continuously monitors water quality and displays live readings on the PWA dashboard using Socket.IO for real-time data streaming from the Node.js backend.

- Live display of water temperature (°C), pH value, and turbidity level
- Configurable reading interval (default: every 10 seconds)
- Historical data graph displaying trends over the past 24 hours, 7 days, and 30 days
- Color-coded status indicators: Green (normal), Yellow (caution), Red (alert)
- Timestamp of the most recent reading displayed on dashboard

6.2 Automated Aerator Control

The aerator is a critical device for maintaining dissolved oxygen levels in the pond. The system provides three modes of aerator control:

- Automatic Threshold Mode: The ESP32 automatically activates the aerator when water temperature exceeds the configured upper threshold (default: 32°C). The aerator runs for a user-configured duration and then stops.
- Scheduled Timer Mode: The user configures up to 3 time-of-day schedules through the PWA. On each schedule, the aerator runs for a configured number of seconds.
- Manual Mode: The user can turn the aerator ON for any configured duration via the PWA. The command is sent through the Node.js API and relayed to the ESP32 via MQTT.

6.3 Automated Fish Feeder Control

- Scheduled Timer Mode: The user configures the exact time(s) of day the feeder should activate (up to 5 daily schedules).
- Duration Control: For each scheduled activation, the user configures how many seconds the feeder motor should run.
- Manual Mode: The user can trigger the feeder instantly from the PWA for a configured number of seconds.
- Status Feedback: The PWA shows whether the feeder is currently running and the time of the last feeding event.

6.4 Alert & Notification System

The alert system operates at two levels simultaneously:

- Local Alert (on-site): The buzzer sounds and the LED flashes whenever any monitored parameter crosses its configured threshold. This occurs entirely on the ESP32 without requiring internet connectivity.
- Web Push Notification: The Node.js backend sends a Web Push Notification to the registered PWA client. The notification message includes the parameter name, current value, and the threshold that was

exceeded (e.g., 'ALERT: Water Temperature is 34°C, exceeding the 32°C threshold'). Push notifications work even when the PWA is not actively open on screen.
- Notification Cooldown: A minimum 10-minute cooldown is enforced between repeated alerts for the same parameter to prevent notification flooding.
- Alert Log: All alerts are logged with timestamp and parameter value in the PWA for review.

6.5 User Configurable Settings

Through the Progressive Web Application, users can configure all of the following settings without any technical knowledge:

- Temperature alert threshold (upper and lower limits)
- pH alert threshold (upper and lower limits)
- Turbidity alert threshold
- Aerator auto-trigger temperature threshold and auto-run duration
- Aerator daily schedule (up to 3 time slots)
- Feeder daily schedule (up to 5 time slots) and run duration per schedule
- Manual aerator and feeder run durations
- Sensor reading interval

6.6 Offline Resilience

- The ESP32 stores the last known configuration (thresholds, schedules) in non-volatile flash memory (NVS). Schedules and threshold-based automation continue to operate offline.
- Local alerts (buzzer and LED) function entirely without internet connectivity.
- Data is buffered locally and published to the MQTT broker once connectivity is restored.

7. Application Overview

7.1 Application Screens

- Home Dashboard: Real-time display of all three sensor readings with color-coded status. Shows aerator and feeder status. Quick-access manual control buttons.
- Historical Data Screen: Line graphs for temperature, pH, and turbidity over selectable time ranges (24h, 7 days, 30 days).
- Aerator Control Screen: View current status. Toggle manual on/off with duration input. View/edit scheduled times and durations. Configure auto-trigger threshold.
- Feeder Control Screen: View last feeding time. Trigger manual feed with duration input. View/edit up to 5 daily scheduled feeding times and durations.
- Alert Settings Screen: Configure threshold values for all three parameters. View alert history log.
- Notification History Screen: List of all past push notifications with timestamps.
- Device Settings Screen: Wi-Fi configuration, sensor calibration offset input, reading interval setting.

7.2 Application Platform

The application is built as a Progressive Web Application (PWA) using React and Tailwind CSS. This approach was chosen for the following reasons:

- Cross-Platform: A single codebase runs on Android, iOS, and desktop browsers without separate native apps.
- Installable: Users can install the PWA directly from the browser (Chrome, Safari) onto their home screen — no Play Store or App Store required.

- Offline Support: Service Workers enable offline caching of the dashboard and last-known readings.
- Web Push Notifications: The PWA receives real-time alerts via the Web Push API, even when the app is not open.
- No App Store Dependency: Updates are deployed to the server instantly, users always receive the latest version automatically.
- Backend Connectivity: The PWA connects to the Node.js/Express.js backend via REST API and Socket.IO for real-time data streaming.

8. Total Cost Estimation

Item	Unit Price (BDT)	Notes
ESP32 Development Board (30-pin)	430	Main controller
DS18B20 Waterproof Temperature Sensor	220	Water temperature
pH Sensor Module (PH4502C)	2,900	pH measurement
Turbidity Sensor Module	820	Water clarity
2-Channel Relay Module	400	Controls aerator & feeder
Buzzer + LED Indicator	200	Local alert
18650 Battery + TP4056 Charger	100	Power supply
Aquarium Aerator (small)	~300	Oxygen pump
DC Feeder Motor	~200	Feed dispenser
Supporting (wires, breadboard, box, resistors)	~200	Miscellaneous
TOTAL ESTIMATED COST	**5,570 - 6,000**	**Base configuration**

Note: Software development uses open-source tools (Node.js, MongoDB Community, Mosquitto). No licensing cost is required for the base configuration. Prices may vary by supplier and availability.

9. Future Expansion Roadmap

The system has been architecturally designed for scalability. The following features are planned for future versions:

- AI-Based Prediction: Machine learning model to predict water quality deterioration based on historical trends stored in MongoDB.
- Multi-Pond Management: Support for multiple ESP32 nodes (ponds) under a single Node.js backend and PWA dashboard.
- SMS Alert System: Integration with a GSM SIM800L module (BDT 350-450) or an SMS API to send alerts even without internet.
- Solar Power Integration: Solar panel (BDT 500-700) and rechargeable battery for fully autonomous outdoor deployment.
- Water Quality Report Export: Weekly/monthly PDF report generation from MongoDB historical data, emailed to the pond owner.
- Fish Health Monitoring: Dissolved oxygen sensor and ammonia sensor additions to the sensor array.

- Commercial Scale: LoRa long-range communication (BDT 600-900) for large ponds where Wi-Fi does not reach.

10. User Manual

10.1 Getting Started

Before using the system for the first time, complete the following setup steps:

- Open the Fish Pond Manager PWA URL in Chrome (Android) or Safari (iOS). Tap 'Add to Home Screen' or 'Install App' to install it like a native application.
- Register an account in the PWA. Log in using your credentials (secured by JWT authentication).
- Power on the ESP32 unit by connecting it to a 5V USB power adapter or the battery system.
- Open the PWA and navigate to Device Settings. Enter your Wi-Fi network name (SSID) and password. The ESP32 will connect to the MQTT broker automatically.
- Once connected, the app dashboard will begin displaying live sensor readings within 30 seconds.
- Calibrate the pH sensor: navigate to Device Settings > pH Calibration. Follow the on-screen instructions using pH 7.0 and pH 4.0 buffer solutions.
- Configure alert thresholds by navigating to Alert Settings. Adjust temperature, pH, and turbidity thresholds to match your fish species requirements.
- Set up feeding schedules by navigating to Feeder Control > Schedule. Add feeding times and durations.

10.2 Reading the Dashboard

- Temperature Card: Displays current water temperature in °C. Background is green when normal, yellow when approaching threshold, red when threshold is exceeded.
- pH Card: Displays current pH value. Green for 6.5-8.5, yellow for 6.0-6.5 or 8.5-9.0, red outside this range.
- Turbidity Card: Displays turbidity reading. Green for low (clear water), yellow for moderate, red for high turbidity.
- Aerator Status: Shows ON/OFF status and the remaining duration if running on a timer.
- Feeder Status: Shows the last feeding time and the next scheduled feeding time.

10.3 Manual Control

To manually run the aerator:

- Navigate to Aerator Control in the PWA.
- Set the desired run duration in seconds using the duration input field.
- Tap 'Run Now'. The command is sent to the Node.js backend via REST API, which publishes an MQTT command to the ESP32. The aerator will run for the specified duration and stop automatically.

To manually trigger the feeder:

- Navigate to Feeder Control in the PWA.
- Set the desired run duration in seconds.
- Tap 'Feed Now'. The feeder motor will run for the specified duration.

10.4 Configuring Schedules

- Navigate to the respective control screen (Aerator Control or Feeder Control).
- Tap 'Add Schedule'. Select the time of day (hour and minute) for the scheduled activation.
- Enter the run duration in seconds. Tap 'Save'. The schedule is sent to the Node.js backend, stored in MongoDB, and synced to the ESP32 via MQTT.

- To delete a schedule, swipe left on the schedule item and tap 'Delete'.

10.5 Understanding Alerts

When an alert is triggered:

- A Web Push Notification appears on your phone with the alert details, even if the PWA is not actively open.
- The local buzzer sounds and the LED flashes at the pond location.
- The affected sensor card on the dashboard turns red.
- The alert is logged in the Notification History screen.

To clear the local buzzer/LED alarm manually, tap 'Acknowledge Alert' in the PWA. This silences the local alarm until the condition returns to normal range.

10.6 Troubleshooting

- No sensor readings on dashboard: Check that the ESP32 is powered and connected to Wi-Fi. Verify that the Node.js backend server is running and the MQTT broker (Mosquitto) is active.
- pH readings seem inaccurate: Recalibrate the pH sensor. Go to Device Settings > pH Calibration.
- Aerator not responding to commands: Check the relay module wiring and verify the MQTT broker is relaying commands to the ESP32.
- Feeder not dispensing at scheduled time: Verify the ESP32's NTP time sync. Ensure the ESP32 has internet access for NTP synchronization.
- App not receiving notifications: Ensure notification permissions are enabled for the PWA in your browser/phone settings. Verify the Web Push subscription is active in Device Settings.

11. Safety & Compliance

- The ESP32 unit must be housed in a waterproof enclosure (IP65 or higher rating) when deployed near water.
- All 240V AC electrical connections must be performed by a qualified electrician. The relay module handles the switching, but AC connections are the user's responsibility.
- Do not submerge the ESP32 unit, relay module, or any electronics. Only the waterproof sensor probes are rated for water contact.
- Ensure the power supply is protected from moisture and rain in outdoor deployments.
- Disconnect power before making any changes to wiring connections.

12. Technical Specifications Summary

Specification	Value
Microcontroller	ESP32 (Dual-core Xtensa LX6, 240 MHz, 4MB Flash)
Connectivity	Wi-Fi IEEE 802.11 b/g/n (2.4 GHz)
Device Protocol	MQTT over Wi-Fi (PubSubClient / AsyncMqttClient library)
Temperature Range	-55°C to +125°C (DS18B20)
Temperature Accuracy	±0.5°C
pH Range	0 to 14 pH (PH4502C)

ADC Resolution	12-bit (0-4095 raw value)
Relay Channels	2 (Aerator + Feeder)
Max Relay Load	10A / 250VAC per channel
Alert Outputs	Buzzer (local) + LED (local) + Web Push Notification (PWA)
Backend Runtime	Node.js + Express.js
MQTT Broker	Mosquitto (lightweight, self-hosted)
Real-Time Sync	Socket.IO
Database	MongoDB (Atlas or self-hosted)
Authentication	JWT (JSON Web Tokens) + Firebase Auth
App Platform	Progressive Web App (React) — installable on Android & iOS, no Play Store needed
Sensor Reading Interval	10 seconds (configurable, min: 5s, max: 60s)
Data Retention (Cloud)	30 days (configurable)
Max Feeder Schedules	5 per day
Max Aerator Schedules	3 per day
Power Supply	5V USB (indoor) / 18650 battery / Solar (outdoor)
Operating Temperature	0°C to 50°C (electronics enclosure)

End of Project Specification & User Manual

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